First-Principles Synthetic OD Generation

Latent Gravity, Heavy Tails, and Constrained Sampling

Implementation: `src/data/first_principles/`, `generate_fp_od_dataset.py`

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Abstract

This document specifies a *first-principles* synthetic origin–destination (OD) generator that does **not** require a historical OD survey matrix. The process samples low-dimensional latent production and attraction factors, builds a gravity interaction kernel with distance decay and approximate reciprocity, applies Gamma heavy-tailed corridor weights, draws Dirichlet compositional noise, scales demand to a fixed total G, and balances marginals with iterative proportional fitting (IPF). Dataset-level quality filters enforce diversity, sparsity, and trip-length realism. The pipeline is implemented separately from the base-OD Dirichlet generator under `src/data/first_principles/`. Turning counts reuse the existing Phase 2 assignment pipeline. We detail each stage, distributional choices, alternatives considered, and pros/cons.

Contents

Chapter 1

Motivation and Problem Framing

1.1 Why first principles?

Machine-learning OD estimators need large labeled datasets. Two strategies exist:

1. **Perturb a survey OD** (existing pipeline): Dirichlet around $T^{\text{base}} + \varepsilon(d)$, then IPF.
2. **Synthesize from network geometry and latent demand** (this pipeline): no survey matrix required.

The second strategy matches the problem statement in `od_generation_problem.pdf`: inputs are TAZs, distances, total demand G, and sample count K only. It also forces the generator to encode transportation structure explicitly (gravity, reciprocity, latent land-use) rather than inheriting it from one observed table.

Table 1.1: Existing vs. first-principles generators.

Aspect	**Base-OD pipeline**	**First-principles (this doc)**
Primary prior	$T^{\text{base}} + \varepsilon(d)$	Latent $u_i v_j f(d_{ij})$
Needs survey OD?	Yes	No
Corridor identity	Anchored near base	Varies with latents + Gamma
Reciprocity	Implicit in base	Explicit mix toward $S^\top$
Spatial coherence	Via ε + base	Spatial smoothing of u, v
Code path	`generate_od_dataset.py`	`generate_fp_od_dataset.py`

1.2 Constraint philosophy

Not every constraint in the problem statement is a hard equality. We partition them as follows.

Table 1.2: Hard vs. soft vs. dataset-level constraints.

Type	**Constraints**	**Enforcement**
Hard (per matrix)	$T_{ij} \geq 0$, $T_{ii} = 0$, $\sum T = G$, P–A consistency, connectivity support	Mask, scale, IPF
Soft (distributional)	Distance decay, heavy tails, reciprocity, spatial coherence, corridor dominance, low-rank latent structure	Latent gravity + Gamma + Dirichlet
Dataset-level	Diversity, entropy, stationarity, sparsity band, trip-length realism	Oversample + quality filters

Trying to enforce reciprocity, freely varying corridors, and arbitrary asymmetric marginals as *exact* hard constraints simultaneously is over-specified. Soft enforcement in expectation is the scientifically correct reading of the problem note.

Chapter 2

End-to-End Process

2.1 Pipeline overview

First-principles OD pipeline

Zones + d_{ij} + $G \xrightarrow{\text{A}}$ latent $(\mathbf{u}, \mathbf{v}) \xrightarrow{\text{B}}$ gravity $S_{ij} = u_i v_j f(d_{ij})$ + reciprocity $\xrightarrow{\text{C}}$ Gamma weights $\xrightarrow{\text{D}}$ Dirichlet $\xrightarrow{\text{E}}$ scale + IPF $\xrightarrow{\text{F}}$ quality filters $\rightarrow \{\mathbf{T}^{\text{syn},(k)}\}$

Reuse Phase 2: $\mathbf{T}^{\text{syn},(k)} \rightarrow \mathbf{x}^{(k)} = \mathbf{A}_{\text{turn}} \mathbf{y}^{(k)}$

Figure 2.1: Stages A–F for OD synthesis; turning counts reuse the existing assignment pipeline.

Notation. N zones; G total trips; d_{ij} inter-zone distance; $\mathbf{u}, \mathbf{v} \in \mathbb{R}_+^N$ production/attraction propensities; $f(d)$ impedance; $\mathbf{S}$ gravity seed; α Dirichlet concentration; a_γ, b_γ Gamma shape/scale; ρ reciprocity mix.

2.2 Algorithm (one replicate)

Algorithm FP-1: First-principles OD (one sample k)

1. Sample type intensities and zone memberships; form $\mathbf{u}^{(k)}, \mathbf{v}^{(k)}$; spatially smooth
2. $S_{ij} \leftarrow u_i v_j f(d_{ij})$ on $i \neq j$; mix $S \leftarrow (1-\rho)S + \rho S^\top$
3. $w_{ij} \sim \text{Gamma}(a_\gamma, b_\gamma)$; $S_{ij} \leftarrow w_{ij} S_{ij}$
4. $\mathbf{p} \sim \text{Dir}(\alpha \, \text{normalize}(\mathbf{S}))$
5. Seed $G\mathbf{p}$; targets $P \propto \mathbf{u}$, $A \propto \mathbf{v}$ (or equal shares); $\mathbf{T}^{\text{syn}} \leftarrow \text{IPF}(\cdot)$

Figure 2.2: Single-matrix generator (`generate_one_fp_od`).

Chapter 3

Stage A: Latent Production and Attraction Factors

3.1 Model

Real OD tables are low-rank in spirit: a few land-use / activity types (residence, employment, retail, ...) drive zone importance. We therefore sample:

1. City-level type intensities $\boldsymbol{\pi}^{\text{prod}}, \boldsymbol{\pi}^{\text{attr}} \in \mathbb{R}_+^r$,

$$\pi_t^{\text{prod}},\ \pi_t^{\text{attr}} \sim \text{Gamma}(a_f, b_f).$$

2. Soft zone–type memberships $\mathbf{M} \in \mathbb{R}^{N \times r}$ with rows on the simplex, obtained by placing r random type centers in the plane and softmax-weighting zones by distance to centers.

3. Zone factors

$$u_i = \sum_{t=1}^{r} M_{it}\, \pi_t^{\text{prod}}, \qquad v_j = \sum_{t=1}^{r} M_{jt}\, \pi_t^{\text{attr}}.$$

Spatial coherence. Nearby zones should not have wildly unrelated importance. We smooth

$$\mathbf{u} \leftarrow (1-s)\,\mathbf{u} + s\,\mathbf{K}\mathbf{u},$$

where $K_{ij} \propto \exp(-d_{ij}^2/(2\lambda_s^2))$ is a row-normalized Gaussian kernel, and likewise for $\mathbf{v}$. Strength $s \in [0, 1]$ and length λ_s are hyperparameters (defaults $s = 0.5$, $\lambda_s = 400\,\text{m}$).

3.2 Why this design?

- **Low-dimensional manifold:** the map $(M, \pi) \mapsto (u, v) \mapsto S$ has effective dimension $\ll N^2$, matching the problem note's latent-structure requirement.
- **Interpretable:** r plays the role of "number of activity types."
- **No survey OD:** factors are drawn from the network geometry and random centers only.

3.3 Alternatives considered

Table 3.1: Latent-factor alternatives: pros and cons.

Method	**Pros**	**Cons**
Independent $u_i, v_j \sim$ Gamma	Simple	No spatial coherence; no shared types
Low-rank $UV^\top$ Gaussian	Classic	Can be negative; weaker land-use story
Fixed census attributes	Realistic if available	Requires external data (not first principles)
Type centers + Gamma (ours)	Geometry-aware; low-dim; non-negative	Center placement is stochastic; r to tune

Chapter 4

Stage B: Gravity Kernel and Reciprocity

4.1 Gravity interaction

Given $\mathbf{u}, \mathbf{v}$ and impedance $f(d)$,

$$S_{ij} = u_i \, v_j \, f(d_{ij}) \, \mathbb{1}[i \neq j]. \tag{4.1}$$

Supported impedance families:

$$f_{\text{exp}}(d) = \mathrm{e}^{-d/\lambda}, \tag{4.2}$$

$$f_{\text{pow}}(d) = (d + \varepsilon)^{-\gamma}. \tag{4.3}$$

Default: exponential with $\lambda = 500\,\mathrm{m}$ (order of median Sioux Falls inter-zone distance).

Why gravity? It is the classical transportation interaction model: productions, attractions, and distance decay enter multiplicatively. In expectation, longer pairs receive less mass—the soft distance-decay constraint—while remaining strictly positive for finite d.

4.2 Approximate reciprocity

Daily demand is often roughly two-way. We mix

$$S \leftarrow (1 - \rho) \, S + \rho \, S^{\top}, \qquad \rho \in [0, 0.5]. \tag{4.4}$$

Default $\rho = 0.35$. This is *not* hard symmetry: Gamma weights and Dirichlet noise later break exact equality $T_{ij} = T_{ji}$, preserving statistical reciprocity.

4.3 Alternatives

Table 4.1: Impedance and reciprocity alternatives.

Choice	**Pros**	**Cons**
Exponential f	One length scale; smooth	Tail thinner than power law
Power-law f	Heavier long-distance mass	Sensitive near $d \approx 0$
Hard $T = T^\top$	Perfect reciprocity	Conflicts with directed corridors / $P \neq A^\top$ patterns
Soft mix ρ (ours)	Tunable; compatible with IPF	ρ is a knob
No distance term	Maximal freedom	Violates soft distance decay

Chapter 5

Stage C: Heavy-Tailed Corridor Weights

5.1 Model

Gravity alone tends to be too smooth. Empirical OD cells are skewed. We reweight

$$w_{ij} \sim \mathrm{Gamma}(a_\gamma, b_\gamma), \qquad S_{ij} \leftarrow w_{ij} S_{ij} \quad (i \neq j), \tag{5.1}$$

with density

$$f(w) = \frac{1}{\Gamma(a_\gamma) b_\gamma^{a_\gamma}} w^{a_\gamma - 1} \mathrm{e}^{-w/b_\gamma}. \tag{5.2}$$

Default $a_\gamma = 0.5 < 1$: density diverges as $w \to 0^+$ and retains a long right tail $\Rightarrow$ many tiny OD pairs, few dominant corridors. Corridor *identity* changes every sample via independent draws—random corridor dominance.

5.2 Why Gamma?

Table 5.1: Heavy-tail mechanisms.

Method	**Pros**	**Cons**
None	Stable	Over-smooth corridors
Gamma $a < 1$ (ours)	Positive; near-zero mass + long tail; simple	Extra (a, b)
LogNormal	Classic heavy tail	Weaker control near zero
Pareto spikes	Extreme hubs	Can break IPF / trip-length filters
Hard top-k mask	Explicit sparsity	Discontinuous; less probabilistic

Chapter 6

Stage D: Dirichlet Compositional Noise

6.1 Model

After Stages B–C, normalize $\mathbf{S}$ to a probability matrix $\mathbf{p}^{\text{prior}}$ on off-diagonal support and draw

$$\tilde{\mathbf{p}} \sim \text{Dirichlet}\big(\alpha\, \mathbf{p}_{\mathcal{S}}^{\text{prior}}\big). \tag{6.1}$$

Mean $\mathbb{E}[\tilde{p}] = \mathbf{p}^{\text{prior}}$; coordinate variance scales as $1/(\alpha + 1)$. Default $\alpha = 200$ (somewhat more diverse than the base-OD pipeline's $\alpha = 500$, because the gravity skeleton already carries structure).

Why Dirichlet? Trip tables are compositional under fixed G. The Dirichlet is the natural distribution on the simplex. Independent Gaussian cell noise would violate the simplex or require awkward renormalization.

Table 6.1: Compositional noise alternatives.

Method	**Pros**	**Cons**
Dirichlet (ours)	Simplex-correct; one knob α	Alone is not enough for extreme skew (hence Gamma)
Logistic-Normal	Flexible correlations	Heavier; harder to tune
No compositional noise	Pure gravity+Gamma	Less fine-grained diversity

Chapter 7

Stage E: Scaling and IPF

7.1 Hard constraints

Set seed $X_{ij} = G\,\tilde{p}_{ij}$ (plus a tiny floor on support for IPF stability). Marginal targets:

$$P_i = G\,\frac{u_i}{\sum_{i'} u_{i'}}, \qquad A_j = G\,\frac{v_j}{\sum_{j'} v_{j'}}, \tag{7.1}$$

or equal shares if `--equal-marginals`. IPF (Furness) alternates row/column scaling until marginal error $< \tau$ or $t_{\max}$ iterations. Post-process: zero diagonal and off-support.

This enforces:

- nonnegativity (up to numerical tolerance),
- $T_{ii} = 0$,
- $\sum_{i,j} T_{ij} = G$ (to numerical accuracy),
- P–A consistency with the chosen targets,
- connectivity support (full off-diagonal for connected zoning systems such as Sioux Falls).

Table 7.1: Balancing alternatives.

Method	**Pros**	**Cons**
IPF / Furness (ours)	Fast; classical; scalable to 10^6	Needs feasible targets
Entropy maximization	Nice dual	Heavier per sample
No balancing	Cheap	Inconsistent zone totals

Chapter 8

Stage F: Dataset Quality Filters

8.1 Procedure

Generate a candidate pool of size $\lceil \kappa K \rceil$ (default $\kappa = 2$), then accept K matrices using the shared module `quality_filters.py`:

- sparsity band;
- trip-length relative error vs. a *reference* gravity OD (deterministic latent draw, no Gamma/Dirichlet);
- optional correlation band vs. that reference (defaults wide: disabled);
- entropy-oriented ranking;
- pairwise Frobenius diversity (reservoir approximation for scalability).

The reference OD is **not** a survey matrix; it is an internal gravity template used only for filter geometry checks.

Table 8.1: Filter design choices.

Choice	**Pros**	**Cons**
Filters on (ours)	Diversity + realism gates	Needs oversampling; thresholds
No filters	Maximum throughput	Near-duplicates possible
Correlate vs. survey OD	Anchors to reality	Contradicts first-principles inputs
Correlate vs. gravity ref	Geometry-aware without survey	Reference is itself synthetic

Chapter 9

Why This Process Makes Sense

9.1 Mapping to the problem constraints

Table 9.1: Constraint coverage map.

Constraint	**Where enforced**
Nonnegativity / no intrazonal	Support mask + IPF post-process
Fixed total G / P–A	Scale + IPF
Distance decay	Gravity $f(d)$ (soft, in expectation)
Heavy tails / corridor dominance	Gamma weights (vary by sample)
Approximate reciprocity	Soft mix ρ toward $S^\top$
Spatial coherence	Spatial smoothing of u, v
Smooth O/D importance	Latent factors u, v
Diversity / entropy	Dirichlet α + quality filters
Sparsity	Gamma + sparsity filter band
Connectivity	Off-diagonal / connected support
Stationarity	Shared process + filters
Low-dimensional latent structure	Rank-r type model
Scalability	Multiprocessing + vectorized filters

9.2 Scientific rationale

The generator is a *hierarchical probabilistic model*:

$$(\mathbf{u}, \mathbf{v}) \to \mathbf{S}_{\text{grav}} \to \mathbf{S}_{\text{heavy}} \to \mathbf{p} \to \mathbf{T} = \text{IPF}(G\mathbf{p}).$$

Each arrow adds a transportation mechanism that the ML inverse problem should eventually recover from turning counts. Because structure is injected by geometry and latents—not by cloning a single survey—the dataset explores a realistic manifold of OD tables while remaining on-policy for the problem statement.

Chapter 10

Implementation and Usage

10.1 Module layout

Table 10.1: Code map (`src/data/first_principles/`).

Module	Role
config.py	FPGeneratorConfig
latent.py	Stage A: types, memberships, spatial smooth
gravity.py	Stage B: impedance, gravity, reciprocity
heavy_tails.py	Stage C: Gamma weights
compositional.py	Stage D: Dirichlet
generator.py	Stages A–E + parallel batch
generate_fp_od_dataset.py	CLI + Stage F filters + I/O
generate_turning_counts.py	Existing Phase 2 (unchanged)

The legacy pipeline under `generate_od_dataset.py` is untouched.

10.2 Example commands

Generate OD matrices.

```
python -m src.data.generate_fp_od_dataset \
  --samples 100000 --oversample-factor 2 \
  --total-demand 365475 --latent-dim 4 \
  --alpha 200 --reciprocity 0.35 \
  --gamma-shape 0.5 --lambda-decay 500 \
  --output-dir outputs/od_generator_fp \
  --output-name synthetic_od_fp.npy \
  --memmap-save
```

Turning counts (existing pipeline).

```
python -m src.data.generate_turning_counts \
  --network sioux-falls.net.xml \
  --od outputs/od_generator_fp/synthetic_od_fp.npy \
  --output-dir outputs/turning_counts_fp \
  --skip-plots
```

10.3 Default hyperparameters

Table 10.2: Recommended defaults (Sioux Falls scale).

Symbol / flag	**Default**	**Role**
G / `--total-demand`	365475	Fixed trip total
r / `--latent-dim`	4	Activity types
λ / `--lambda-decay`	500	Distance decay length (m)
ρ / `--reciprocity`	0.35	Symmetry mix
a_γ / `--gamma-shape`	0.5	Heavy-tail shape
α / `--alpha`	200	Dirichlet concentration
κ / `--oversample-factor`	2	Filter pool size

Chapter 11

Limitations and Future Work

- Membership centers are geometric proxies, not calibrated land-use layers.
- Reciprocity is a linear mix, not a structural home–work matching model.
- Connectivity currently uses full off-diagonal support (valid for fully connected zone systems); sparse network cut-sets could be added.
- Hyperparameters $(r, \lambda, \rho, \alpha, a_\gamma)$ still need network-specific calibration via held-out realism metrics.
- Optional: replace Euclidean d_{ij} with network shortest-path costs for impedance.

Conclusion

The first-principles pipeline generates OD matrices from zones, distances, and total demand alone by combining latent land-use factors, gravity interaction, Gamma corridor noise, Dirichlet compositional sampling, IPF balancing, and dataset filters. Each stage targets a subset of the constraints in `od_generation_problem.pdf`. The implementation lives beside—not instead of—the existing base-OD generator, and turning counts reuse Phase 2 unchanged.

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